

Question:
A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

- Answer Choices:
- A. Epstein-Barr virus (EBV)
 - B. Measles virus
 - C. Human immunodeficiency virus (HIV)
 - D. Human papillomavirus

Reasoning + Answer:

<think> B cells that do not express CD21 will be unable to bind with the virus that uses complement component 3 which is a coreceptor for the virus and this will specifically affect the entry of the Epstein-Barr virus into the B cells and replicate and cause infectious mononucleosis so removing cd21 from the B cells will stop the virus from binding and infecting the B cells and the virus is dependent on binding to complement component 3 and cd21 to enter the B cells and replicate </think>
<answer> A. Epstein-Barr virus (EBV)</answer>

